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Elite athletes' and support staffs' experiences and perceptions of long-haul travel, and the self-management strategies they use

Antonia Rossiter[1,2,3], Thomas M. Comyns[2,3], Ian Sherwin[2,3], Alan M. Nevill[2,4], and Giles D. Warrington[2,3]

Abstract

Elite athletes and their support staff are often required to travel for international competitions all over the globe, however little is known about their experiences of long-haul (LH) travel and its perceived effects on performance. The aim of this study was to explore elite athletes' and support staffs' perception of symptoms experienced following LH travel and the self-management strategies they used to mitigate these symptoms. Elite athletes and support staff ($n = 88$), who had embarked on an LH flight (> 8 hours) in the previous 2 years for training or competition, completed a survey examining their perceptions of LH travel, symptoms experienced following the flight(s) and strategies they implemented to minimise the symptoms. Associations between symptoms experienced and travel strategies used with participant and journey characteristics were examined by Chi-squared tests. LH travel was widely perceived by participants to be disruptive to physical (86.4%) and mental performance (72.7%) and to increase the risk of illness and injury (86.4%). The most common symptoms experienced were related to fatigue and disruption to sleep. All participants implemented strategies to help mitigate the negative consequences of LH travel. Moving and stretching regularly in-flight and simple strategies for aligning the body clock to destination time were most prevalent. The study findings will allow the translation of research to better inform future guidelines that address the unique needs and priorities of elite athletes and support staff as they embark on LH travel for training and competition with a view to optimising performance outcomes.

Keywords

Chronotypes, circadian rhythms, fatigue, jet lag, sleep, time zones

Introduction

High-performance (HP) athletes and their support staff frequently embark on long-haul (LH) travel for training and major international competitions. LH travel can result in travel fatigue and jet lag, both of which can potentially prevent optimal physical and mental performance.[1–3]

Travel fatigue is caused by challenges of LH travel, such as stresses encountered on the journey,[4,5] sleep disruption before[6,7] and during the journey,[1,8,9] limited access to preferred food/fluid,[10] and the aircraft cabin environment.[4,5] Travel fatigue can be acutely experienced after any individual long journey, in any direction; or chronically because of repetitive travel within a season.[11,12] Recovery from travel fatigue is expected to take a day or two with adequate rest, sleep, and nutrition.[1,4,5,11] However, the 'recovery window' should account for travel distance, travel time, travel frequency and length of the season.[12]

Reviewers: Hugh Fullagar (University of Sydney, Australia)
Shona Halson (Australian Catholic University, Australia)

[1]Sport Ireland Institute, Sport Ireland Campus, Dublin, Ireland
[2]Department of Physical Education and Sport Science, University of Limerick, Limerick, Ireland
[3]Sport and Human Performance Research Centre, Health Research Institute, University of Limerick, Limerick, Ireland
[4]Faculty for Education Health and Wellbeing, University of Wolverhampton, Wolverhampton, UK

Corresponding author:
Antonia Rossiter, Sport Ireland Institute, Sport Ireland Campus, Dublin, D15 Y52H, Ireland.
Email: arossiter@instituteofsport.ie

A recent consensus statement has defined jet lag as a 'temporary impairment of sleep and wakefulness, as well as other biological functions, associated with rapid eastward or westward travel across three or more time-zones'.[12] Jet lag is caused by desynchronisation of the body's circadian system with external environmental cues.[13] The experience of jet lag greatly depends on the direction of travel. After flying eastward, the circadian system will be running behind the local time zone, therefore the circadian nadir will occur in the morning time. Thus, following LH eastward travel a person will feel sleepy in the morning time and they may have difficulty falling asleep at their normal bedtime. An athlete may not perform or compete at their best in the morning time on the days after LH eastward travel. Following westward travel, the circadian system will be running ahead of the local time zone and the circadian nadir will be in the evening time, therefore it is expected that a person may have difficulty staying awake and an athlete may not perform or compete at their best in the evening time.[14] Symptoms of jet lag are generally more severe and last longer following an eastward flight in comparison to a westward flight[14–16] and when a greater number of time zones are crossed.[1,15]

The human body clock needs time to gradually adjust and consequently the symptoms of jet lag will persist until circadian rhythms shift to the new environment.[1,4,16] On arrival to the new time zone, it is imperative that the HP sport team are allowed sufficient time to recover from jet lag so they can optimise their preparation to perform at the highest level. Generally, one day is needed to recover from each time-zone crossed,[17] however, it has been suggested that HP athletes may need longer.[6,18] A study of both athletes and non-athletes reported higher jet lag in athletes and postulated that the daily routine of high workload and busy training schedules may have rendered athletes more prone to the negative effects of LH transmeridian travel.[19] Other possible reasons are inappropriate timing of sleep and exposure to zeitgebers such as light, physical activity and meals.[12]

'Travel' has been identified as a stressor experienced by elite athletes,[20] HP coaches[21] and sport science and management staff in elite sport.[22] LH transmeridian travel has been found to be associated with physical and physiological decrements, cognitive and psychomotor disturbances, and sleep disruption in the general population,[11,15,19,23–25] in elite athletes,[6] and HP support staff.[7] The effects of LH travel on athletic performance are not definitively established, particularly in an applied setting,[3,26,27] however disturbances in athlete physical and mental state, poor sleep and illness are all known to be associated with under-recovery and reduced athlete performance.[28–30]

Recently, here has been increased awareness of guidelines to optimise recovery from LH travel and minimise its negative effect on athlete health, well-being, and performance.[12,14] However there has been little consideration of the coaching and support staff, who experience jet lag, elevated fatigue, sleep, and mood disruption after LH travel,[7] with little or no opportunity to recover from the journey. Despite this, HP support staff are generally required to be in an emotional, physical and mental state to perform at an optimal level on arrival to ensure athletes experience a smooth transition into their new training/competition environment.

A recent scoping review[31] suggested a disconnect between the travellers' lived experiences of travel fatigue and jet lag and the researchers' focus on circadian misalignment. Similarly, a recent review investigating elite athletes concluded that there is currently insufficient understanding of the effects of LH travel on elite athlete performance and the underlying mechanisms.[27] Observational studies on international athletes[6] and Olympic team support staff[7] have identified a potential mismatch between the expected effects of the circadian cycle on sleep schedules following LH transmeridian travel and the actual effects measured in the field. Similar to the general population,[32] this suggests that findings and recommendations based on circadian science do not always easily translate to an applied sport setting.

It has recently been suggested that greater research focus is required to help understand behavioural contexts that drive the successful implementation strategies applicable to performance in elite sport.[33] Little is known about the lived experiences of LH travel from the perspective of HP athletes and support staff, their understanding of jet lag and travel fatigue, the perceived impact that LH travel has on their performance, the strategies (if any) used in association with LH travel for training camps or competition, and what informs the choice of strategies used. It has been reported that there is a high individual variability in the types and severity of jet lag symptoms exhibited and varying rates of recovery following LHTT, generally contributed to characteristics such as sex, age, chronotype, and travel experience.[34] However, it is not known if this variability translates to HP athletes and support staff, and if self-management strategies for LH travel for support staff need to be considered independently.

This study aimed to explore HP athletes' and support staffs' perception of symptoms experienced following LH travel, and to determine the strategies that they used to mitigate these symptoms. This will better inform the design and delivery of targeted interventions for HP athletes and support staff, will optimise recovery and performance after LH travel in a manner that aligns with their unique needs and priorities and considers the specific needs of the support team as well as the athletes.

Methods

Participants

This study was open to all Irish HP athletes, coaches, and support staff, >18 years and competing or working with

athletes at an elite level (professional, Olympic/Paralympic, international, and national). Support staff included sports' chief executive officers, HP directors, physiotherapists, nutritionists, medical doctors, sports psychologists, physiologists, performance analysts, strength and conditioning coaches, athlete lifestyle advisors, and others involved in the daily training environment. Participants were recruited via social media (Facebook and Twitter), posters and by email containing a URL and QR code for the surveys.

Procedure

The surveys were conducted using an online survey platform (Survey Monkey™, Palo Alto, CA, USA) and conformed with General Data Protection Regulation laws. The surveys were open between March 21 and May 31, 2019. The surveys contained up to 44 questions and took approximately 20 minutes to complete and were enabled for completion on any electronic device (i.e. smartphone, computer, or tablet).

All participants were provided with information about the purpose and nature of the survey and provided informed consent before entering the main component of the survey. The survey was voluntary and anonymous. The study was approved by the University's Research Ethics Committee and conformed to the declaration of Helsinki pertaining to research using human subjects.

Survey content

The survey questions were adapted from previous research[5] and questionnaires[35,36] related to the symptoms and management strategies for travel fatigue and jet lag. The initial survey was piloted with an expert panel that consisted of three researchers and four HP support staff from a range of disciplines who had been working in HP sport for at least 6 years and were experienced in LH travel for training and competition. Feedback from the expert panel was positive but they did provide some specific recommendations with regard to the content and terminology of a small number of questions. The final amended survey consisted of two versions (an 'athlete' and a 'staff' version). The majority of questions were the same in sections 1–4 of both versions of the surveys, except for questions in Sections 1 and 3 as outlined below.

Section 1: Background information. Basic demographic details were collected, including participant age and gender. Participants were also asked to identify their chronotypes. Sport-related characteristics included their role in the sport (staff version only) and the type and level of sport(s) they competed, coached, or supported. Participants who did not complete Section 1 were excluded from this study.

Section 2: Experiences and perceptions of LH travel. This section sought to determine the number of LH flights that HP athletes and staff had embarked on in the previous 2 years, and the perceived effects of LH travel on physical and mental performance and risk of injury. Questions in this section were the same in the athlete and staff versions of the survey.

Section 3: Most recent performance-related LH flight. This section focussed on the most recent LH flight that HP athletes and staff embarked on for training and competition in the previous 2 years and the symptoms they experienced in the three days after the flight. This section also addressed strategies that were implemented to reduce the negative effects of the LH flight for training or competition. Strategy implementation was divided into three timeframes (1) pre-travel, (2) in-flight, and (3) post-flight strategies. Most questions in this section were the same in the athlete and staff versions of the survey, except for questions referring pre- and post-travel management strategies, where 'adjustment to training load' used in the athlete version was replaced by 'adjustment to physical activity' in the staff version.

Section 4: Travel strategy education and advice. Participants were asked if they sought advice regarding travel strategies, the reasons behind seeking/not seeking advice and how useful they perceived this advice. Questions in this section were the same in the athlete and staff versions of the survey.

Open responses. For questions regarding symptoms and strategies, participants were given the opportunity to input their own information, outside that captured by the survey.

Data analysis

This study concerns data from a subset of the participants, that is, athletes and support staff who had travelled on an LH flight >8 hours duration for training or competition in the previous 2 years. The relevant athlete and staff data were pooled into one data set, as it offered greater statistical power and more effective subgroup analysis.[37] Categorical variables were summarised using frequencies and percentages, and continuous variables were summarised using mean and standard deviation. Open responses were analysed qualitatively and coded as belonging to an existing symptom or strategy or forming a new symptom or strategy. Associations between symptoms experienced and travel strategies used with participant and journey characteristics were examined by Chi-squared tests. Where >20% of cells had an expected count <5, Fisher's exact test was used instead of the Chi-squared test. All data analyses were conducted in SPSS (Version 28, IBM, Armonk, New York, NY, USA).

Results

Participant characteristics

Overall, there were 144 respondents that satisfied the study inclusion criteria (77 to athlete survey and 67 to staff survey) aged 18 to 66 years (mean age 34.8 years, SD 12.8 years), and 47% were female. This study focused on a subset of 88 HP athletes ($n = 50$) and support staff ($n = 38$) who had embarked on LH travel (flight >8 hours) in the previous 2 years for training and competition. Participant characteristics are displayed in Table 1. Participants were involved across a range of 14 sports (athletics, boxing, clay target shooting, cricket, cycling, diving, golf, gymnastics, hockey, rugby, sailing, skeleton, swimming, and triathlon, Figure 1) and HP levels (professional, Olympic/Paralympic, international, and national). 39.5% of the support staff respondents worked across various sports (Figure 1). 47.7% of participants had embarked on ≥5 LH journeys for training or competition in the past 2 years.

Journey characteristics

All participants experienced a ≥ 1 h time difference between origin and destination (M 6.1 hours, SD 2.5 hours). 88.7% of participants experienced a time difference of >3 hours on arrival at their destination. 59.1% of participants travelled in an eastward direction and 40.9% of participants travelled in a westward direction.

Perceived effects of LH travel on performance and health

Most participants agreed or strongly agreed that LH travel could potentially hinder physical performance/ability (86.4%), mental performance/ability (72.7%) and potentially place them at a greater risk of illness/injury (86.4%) (Table 2).

Symptoms experienced

Figure 2 outlines symptoms experienced by the participants in the 3 days following their most recent flight for training/competition. 93.2% of participants reported experiencing ≥1 symptom in the 3 days after arrival, 5.7% of participants did not report any symptoms and 1.1% of participants could not recall experiencing any symptoms. The most frequently reported symptoms following LH travel were related to fatigue (increased fatigue during the daytime (55.7%) and general weariness (51.1%)) and disruption to sleep (early waking time (51.1%) and difficulty getting to sleep (43.2%)). None of the participants reported experiencing symptoms outside those captured by the questionnaire.

27.3% of participants reported experiencing illness/injury after arrival, with 15.9% participants reporting upper respiratory tract infection (URTI) complaints (sinus issues, sore throat, head cold).

Females more frequently reported experiencing daytime fatigue (71.1% vs. 44.0% in men, $X_1^2 = 6.4$, $p = .011$), loss of appetite (29.0% vs. 8.0% in men, $X_1^2 = 6.7$, $p = .010$) and disorientation (21.0% vs. 6.0% in men, $X_1^2 = 4.5$, $p = .034$). Age was significantly associated with reports of general weariness following LH travel; least likely to be reported by those in the 18–24-year age category ($X_4^2 = 9.9$, $p = .042$). Daytime fatigue was most likely to be reported by evening chronotype (80.0% vs. 56.7% morning type and 42.1% neutral type, $X_2^2 = 7.6$, $p = .022$). Participants who had less experience of LH travel (≤ 10 trips) in the 2 years previous were more likely to report daytime fatigue ($X_2^2 = 10.9$, $p = .004$) and lower limb swelling ($X_2^2 = 7.3$, $p = .026$).

Travel duration was not strongly associated with any symptoms experienced following LH travel. In contrast, the number of time zones crossed were associated with early waking ($X_2^2 = 11.9$, $p = .003$), general weariness ($X_2^2 = 8.7$, $p = .013$) and reduced motivation ($X_2^2 = 6.6$, $p = .037$). Early waking and general weariness were most likely to be reported in those travelling across ≥9 time zones. Those travelling eastwards were more likely to report increased stiffness and/or soreness (53.8% vs. 22.2% in westward travellers, $X_1^2 = 8.8$, $p = .003$), reduced physical ability (51.9% vs. 19.4% in westward travellers, $X_1^2 = 4.3$, $p = .038$) and reduced ability to concentrate (34.6% vs. 8.3% in westward travellers, $X_1^2 = 8.1$, $p = .004$).

All participants perceived being completely recovered from symptoms in ≤7 days, the majority (70.5%) of participants perceived taking ≤3 days to completely recover. Perceived recovery time was not significantly associated with participant characteristics, travel duration or direction. However, the greater number of time zones crossed was associated with a longer recovery time ($X_8^2 = 16.1$, $p = .041$).

Use of travel strategies

85.3% of participants had sought advice in the past regarding travel strategies with performance (60.2%) and well-being (53.4%) as the most frequently cited reasons for seeking this advice. Only 8.0% of participants reported that they 'never' sought advice. Most participants rated their personal knowledge of travel strategies to promote recovery as 'excellent' (10.2%), 'very good' (21.5%) or 'good' (51.1%), and 'strongly agreed' (23.9%) or 'agreed' (48.9%) that there was adequate advice on travel strategies available to them. 61.4% of participants perceived this advice as 'extremely useful' or 'very useful' and everyone agreed that at least 'a little' of the advice received was applied, with 35.3% of participants applying 'a lot' of the advice.

Participants' use of travel strategies pre-travel, in-flight and post-flight are shown in Figure 3 as captured by the survey. The use of travel strategies was high with all participants

Table 1. Participant characteristics.

	Athlete	Staff	All participants
n (%)	50 (58.8)	38 (43.2)	88 (100)
Age (years, M ± SD)	26.5 ± 7.2	47.4 ± 8.8	35.5 ± 13.1
Sex			
Male *n* (%)	24 (27.3)	26 (29.5)	50 (58.8)
Female *n* (%)	26 (29.5)	12 (13.6)	38 (43.2)
Chronotype			
Morning type *n* (%)	16 (32.0)	14 (36.8)	30 (34.1)
Evening type *n* (%)	14 (28.0)	6 (15.8)	20 (22.0)
Neutral *n* (%)	20 (40.0)	18 (47.4)	38 (43.2)
Competitive Level			
Professional *n* (%)	5 (10.0)	7 (18.4)	12 (13.6)
Olympic/Paralympic *n* (%)	14 (28.0)	27 (71.1)	41 (46.6)
International *n* (%)	28 (56.0)	4 (10.5)	32 (36.4)
National *n* (%)	3 (6.0)	0 (0)	3 (3.5)

reporting using at least one strategy post-travel, 96% in-flight and 94% pre-travel. Overall, the most frequently used strategy was ensuring to go to bed (83.0%) and eating meals (72.7%) at appropriate local/destination time on arrival. The most frequently used strategies pre-travel were avoiding alcohol (46.6%) and adjusting fluid intake (44.3%), and during the flight it was moving around and stretching regularly (73.9%) and adjusting watch time to destination time (63.6%). Participants were given the opportunity to report the use of other strategies, and open responses from 10% of the participants tended to reinforce the behavioural strategies captured by the survey. The following strategies were reported by participants, but not captured by the survey; the use of meditation/relaxation (4%), in-flight comfort (cushion/eye mask) (4%), medicated nasal spray (4%) and mobility exercises on arrival (4%).

Participant characteristics affected the timing of strategy implementation. Athletes were more likely to adjust their bedtime (46.0% vs. 13.2% staff, $X_1^2 = 10.7$, $p = .001$) and wake time (34.0% vs. 10.5% staff, $X_1^2 = 6.5$, $p = .011$) pre-travel. Younger participants (18–34-year category) were more likely to adjust pre-travel bedtime ($X_4^2 = 10.3$, $p = .036$), wake time ($X_4^2 = 10.5$, $p = .032$), mealtimes ($X_4^2 = 15.2$, $p = .004$) and adjust mealtimes to destination time on the flight ($X_4^2 = 15.5$, $p = .004$). However, on arrival staff (18.4% vs. 2.0% in athletes, $X_1^2 = 7.1$, $p = .008$) and participants ≥35 years ($X_4^2 = 10.0$, $p = .042$) more likely to avoid light.

There was no significant association between travel duration and strategies used before or after LH travel, however >70% of those embarking in longer travel duration (≥15 hours) reported changing their watch time to destination time on the flight in comparison to travel duration of ≤14 hours (40.9%, $X_2^2 = 6.6$, $p = .036$). Those who were crossing ≥9 time zones were most likely to adjust their sleep schedule to destination time on the flight: bedtime ($X_2^2 = 7.3$, $p = .026$) and wake time ($X_2^2 = 13.2$, $p = .001$). They were also more likely to seek bright light on arrival ($X_2^2 = 13.4$, $p = .001$). There was no significant association with direction of travel and strategy-use pre-travel but on the flight those embarking in eastward travel were more likely to change their watch to destination time (73.1% vs. 50.0% westward, $X_1^2 = 4.9$, $p = .027$) and adjust their sleep schedule to destination time; bed time (63.5 vs. 36.1% westward, $X_1^2 = 6.4$, $p = .012$), wake time (40.4% vs. 16.7% westward, $X_1^2 = 5.6$, $p = .018$). On arrival, those travelling eastward were also more likely to ensure to eat meals at an appropriate time (82.7% vs. 58.3% westward, $X_1^2 = 6.3$, $p = .012$) and seek bright light (46.2% vs. 22.2% westward, $X_1^2 = 5.3$, $p = .022$).

Discussion

The results of the survey provide valuable insight into the symptoms experienced by Irish HP athletes and support staff following LH travel for training and competition and the self-management strategies they apply to reduce the negative consequences of LH travel. The majority of this study's participants, who had all experienced LH travel for training and competition, agreed that LH travel could potentially hinder physical and mental performance and increase the risk of illness/injury and 93% of participants had experienced at least one negative symptom following their most recent LH trip for training and competition. The Irish HP sport community generally perceived themselves as knowledgeable (73%) and well-supported (83%) with regard to strategies to help with the management of LH travel. 85% of this study's participants had previously sought advice to help manage the negative effects of LH travel. And all participants reported applying at least one strategy following their most recent LH flight for training and competition. The current scientific literature suggests that one day is needed to recover from each time zone

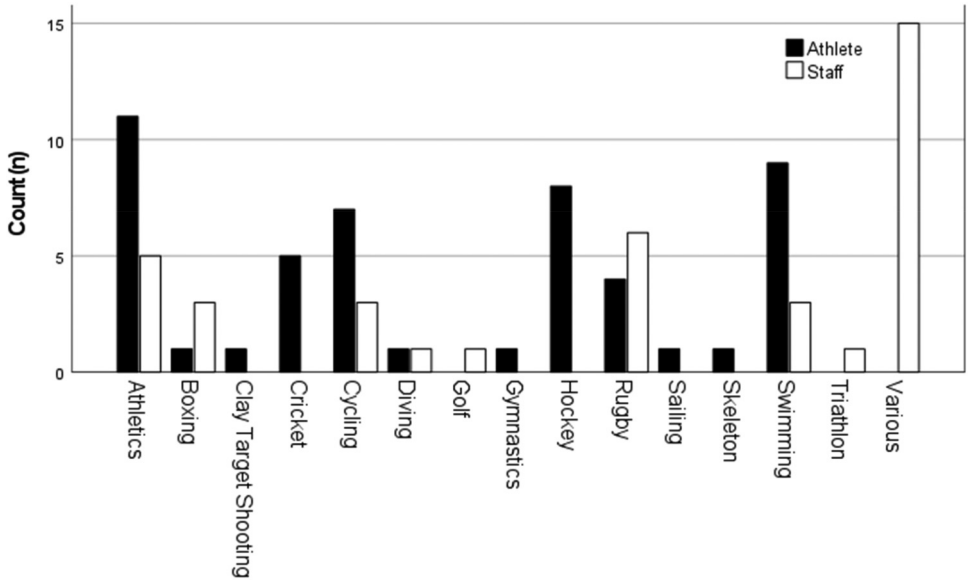

Figure 1. Proportion (*n*) of participants by role (athlete or staff) and sport ($N = 88$).

Table 2. Participant perception of long-haul travel on physical and mental performance/ability and risk of injury.

	Reduced physical performance/ ability *n* (%)	Reduced mental performance/ ability *n* (%)	Greater risk of illness/ injury *n* (%)
Strongly agree	35 (39.8)	23 (26.1)	36 (40.9)
Agree	41 (46.6)	41 (46.6)	40 (45.5)
Neither agree nor disagree	10 (11.4)	17 (19.3)	8 (9.1)
Disagree	2 (2.3)	6 (6.8)	3 (3.4)
Strongly disagree	0 (0)	1 (1.1)	1 (1.1)

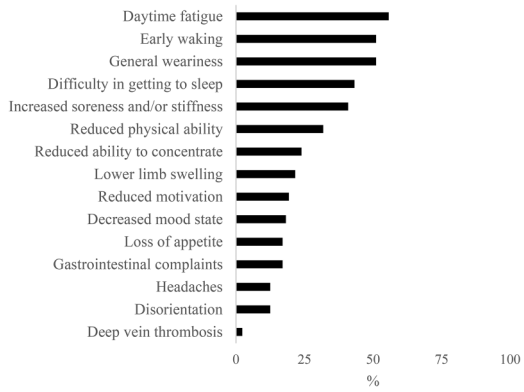

Figure 2. Percentage (%) of participants experiencing symptoms associated with long-haul travel in the 3 days following the most recent flight for training or competition.

crossed[15,24] but depends on the number of time zones crossed, the direction of travel and individual factors.[1,14] The current study showed that although greater number of time zones crossed was associated with longer recovery time, all participants considered themselves completely recovered from symptoms within 7 days, with most participants fully recovered within 3 days. These findings suggest that the Irish HP sport community perceive themselves as susceptible to the negative effects of LH travel, however, are highly motivated to seek advice and apply strategies to reduce the consequences of LH travel. More rapid recovery times than anticipated following LH travel suggest that the self-management strategies implemented by this population are effective in improving recovery from LH travel.

In agreement with studies on the general population,[35,38] international swimmers,[6] and HP support staff,[7] the most prevalent symptoms reported by HP athletes and support staff in the 3 days following LH travel for training/competition were fatigue-related symptoms and disruption to sleep. Sleep disruption before,[6,7,39] during,[40,41] and after[6,7] LH travel for training and competition is likely to have contributed to elevated fatigue after travel. Sleep disruption is associated with LH travel for a myriad of reasons, including flight scheduling,[42] air cabin conditions and seating,[8,43] unfamiliar sleep environment,[44,45] training[42] and work schedules,[7] and pre-competition nerves.[40,46] Furthermore, travelling across time zones can result in the body's internal drive to sleep and wakefulness being out of sync with the destination time[5,16] and can also result in

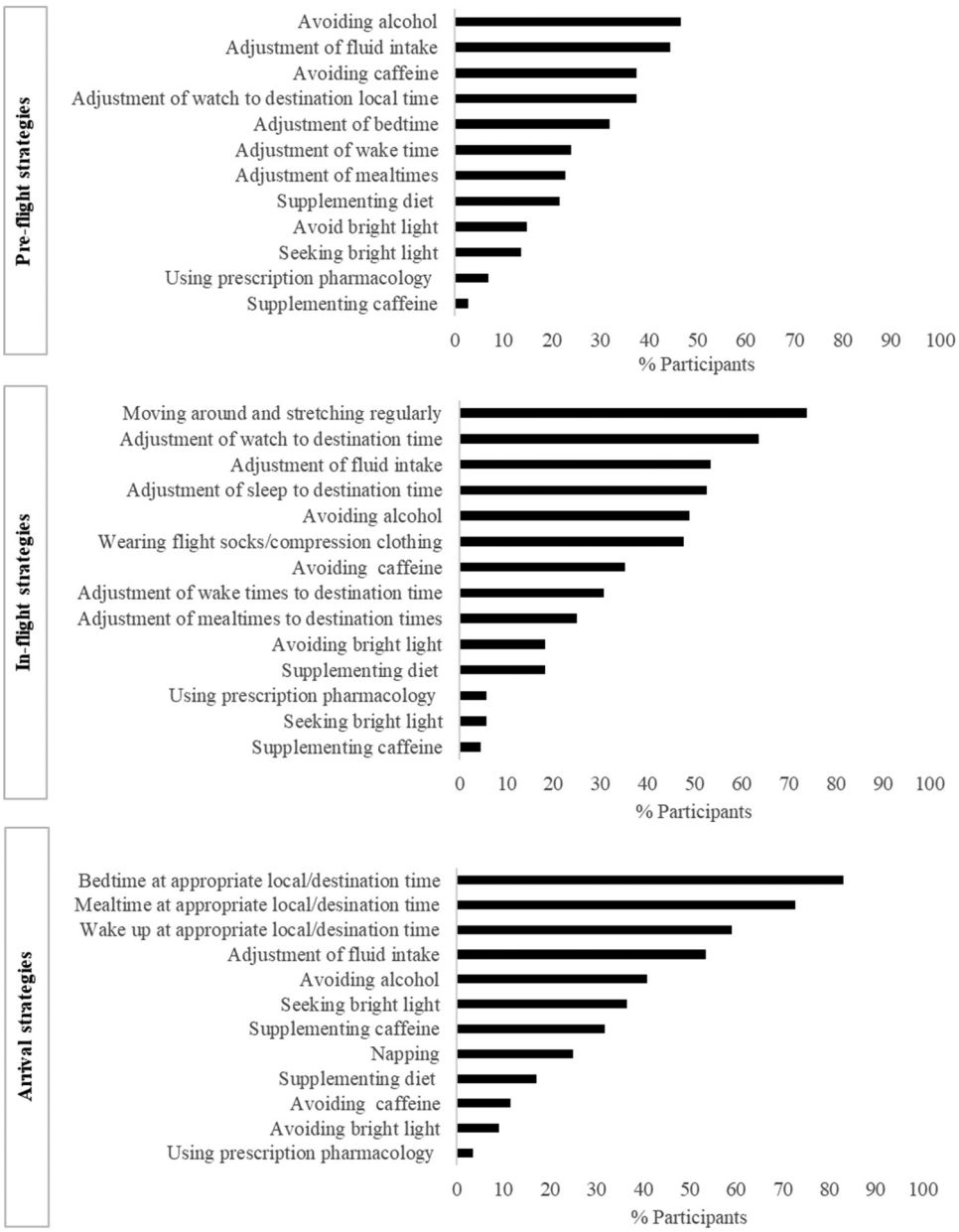

Figure 3. Percentage (%) of participants implementing strategies before, during and after long-haul travel for training or competition.

daytime fatigue and weariness due to the presence of the nadir of the core body temperature during the day.[38] The current study found that early waking and general weariness were more frequently reported by participants travelling across a greater number of time zones. Evidence suggests that poor sleep and elevated fatigue are consistently predictive of perceived jet lag.[35,47] However, a relatively high proportion of the participants travelling ≤ 3 time zones also experienced difficulties getting to sleep (50%) and general weariness (50%) following LH travel. These findings are supported by a study of professional soccer players examining the effect of northbound LH travel that did not involve crossing time zones.[48] The authors showed that athletes reported feeling 'jet lagged' and experienced disruption to perceptual markers such as mood, motivation, perceived stress, and fatigue. This indicates that disruption to sleep and fatigue can occur with LH travel, even with minimal disruption to circadian rhythm, and should be the focus of attention of future strategy development, regardless of the number of time zones crossed. These study findings suggest that future travel strategies such as scheduling of work, training, and flights to minimise sleep disruption and recommendations on strategic napping and sleep hygiene may be countermeasures to sleep disruption encountered before, during and after LH travel and daytime fatigue on arrival.

Despite sleep disruption and daytime fatigue being the most frequently reported symptoms following LH travel, strategies to assist with the alleviation of daytime fatigue on arrival such as napping were less frequently used. The participants in this study more frequently selected self-management strategies that might be perceived as facilitating alignment of their body clock with their destination time, such as adjusting their watch to their destination time on the flight and ensuring that they ate meals and went to bed at times appropriate to their destination on arrival. These findings suggest that the current participants selected easily accessible self-management strategies at a stage of the journey that best facilitated their specific performance requirements. This is supported by the literature focussing on LH travel for professional purposes. It has been found that international business travellers deliberately choose flights that arrive at their destination at a particular time.[49] Unsurprisingly given the safety implications, pilots and airline staff have been found to focus on sleep strategies that minimise sleep loss before travel[50] and optimise alertness during travel.[51]

Despite most of the existing literature suggesting that appropriately timed exposure to light and ingestion of exogenous melatonin before, during and after an LH flight may aid successful circadian realignment,[38,52,53] less than 7% of participants in the present study reported using prescription pharmacology such as melatonin and sleep aids. Less than 20% reported actively seeking or avoiding light before or during travel, and this increased to 38% of participants reporting seeking bright light after travel. It is possible that the light exposure/avoidance regimen might be too restrictive and impractical to implement around training times or work commitments. There may be concern amongst this population that the mistimed or inappropriate exposure to light or use of pharmacological agents may come with the risk of detrimental side effects.[25,52] Furthermore, these strategies have not always been found effective for the alleviation of jet lag symptoms in elite athletes.[54] Recent surveys have reported the use of sleep aids and melatonin at 15% and 8% respectively in Australian airline passengers[31] and a little higher for Australian international business travellers at 19% for sleep medication and 17% for melatonin.[49] The low rate of pharmacological usage found amongst the Irish HP athletes and staff may be due to reluctance to use such medications for fear of breaching WADA 'banned substance' rules.

The current study found increased stiffness/soreness following the flight was experienced by 41% of participants and was most frequently reported by those travelling eastwards. External physical factors on an LH flight, such as confined and uncomfortable space for a prolonged period, restricted movement, muscle inactivity and vibration effects, may contribute to reduced physical performance[12] and increased stiffness and soreness after the flight. Peak physical performance (including joint flexibility) occurs between 16:00 and 22:00.[55] Immediately after flying west, the circadian clock will be running ahead of the local time, therefore peak physical performance is still likely to occur during the day. However, after flying east, the circadian clock will be running behind the local time, therefore the circadian time for peak physical performance is likely to be at night-time and the nadir for physical performance is likely to occur during the daytime. It has been reported that airline travellers rely on obvious symptoms as cues to initiate behaviour change.[32] Considering the high prevalence of soreness/stiffness following the flight it is unsurprising that the most frequent in-flight strategy reported by the participants in this study was moving around and stretching regularly on the flight. These study findings suggest that future travel strategies should include regular movement around the air cabin during the flight when permitted and light mobility routines before, during and after the flight. Recommendations around the scheduling of the athlete's key training sessions so that they can deliver peak physical performance on arrival to the competition/training destination may also be appropriate.

Most participants in this study (86%) perceived LH travel to be associated with an increased risk of illness/injury, with over 27% of participants self-reporting illness/injury within 3 days following their most recent LH travel for training and competition. 16% of participants reported URTI complaints (sinus issues, sore throat, head cold). Other illnesses and injuries reported were gastroenteritis and back/hip pain. It has been suggested that up to 20% of airline passengers may develop symptoms of URTI within one week of airline travel.[56] A greater incidence of illness, including URTI, has been reported in elite athletes following LHTT.[57–59] The enclosed environment of a commercial aircraft cabin exposes passengers to hypobaric hypoxia, dry humidity, and proximity to fellow passengers,[60] thus is a high-risk environment for transmission of infectious diseases such as URTI. Compared to sedentary populations, athletes may be at an increased risk of illness following international travel because of compromised immune function due to intensive training.[61] It is recommended that HP athletes and support staff continue to practice the strategies developed during the COVID-19 pandemic to reduce the risk of contracting illness while travelling.

The current study found that participant characteristics were associated with some symptoms reported following LH travel. Female participants more frequently reported specific symptoms, such as daytime fatigue, GI disturbance and disorientation. In previous studies, there is little evidence that gender influences jet lag.[13,35] However, in this instance the method of data collection may have influenced symptom reporting with surveys consistently showing females to have higher symptom reports than males,

particularly when healthy subjects are concerned, and when symptoms are measured in retrospect.[62] The present study showed evening chronotypes (80%) were more likely to report daytime fatigue in comparison to morning (57%) and neutral (42%) chronotypes. Evening types prefer going to bed later, waking up later and being active in the evening and research has found that they may be more at risk of being sleep deprived than morning types if they rise earlier than their natural preference.[63] Similarly, in the current study, if the evening types did not adjust their bedtime but woke earlier following LH travel this may put them at greater risk of shortened sleep duration and thus increased daytime fatigue. The youngest age group (18–24 years) was least likely to report symptoms of general weariness and GI complaints. Athletes were also less likely than staff to report GI complaints, however, the prevalence of all other symptoms were similar between athletes and support staff. There is contrasting evidence as to whether increasing age is a risk factor for jet lag symptoms.[13] In a study of 85 athletes, academics, and coaches travelling eastward across 10 time zones, older subjects experienced fewer jet lag symptoms, including less fatigue, than younger subjects, however, there were no significant predictors for bowel activity.[19] The present study found that participants who had embarked on more LH trips in the previous 2 years were less likely to experience daytime fatigue and lower limb swelling. The reasons for this are unclear but it may be a reflection of their ability to effectively implement self-management strategies to avoid such symptoms. These findings are supported by a study on professional soccer players which found the number of international appearances, which might be assumed to indicate greater travel experience, were associated with reduced jet lag and improved function on the first day after travel.[64]

Participant and journey characteristics did not largely affect the implementation of travel strategies. Given that the participants in this study were relatively well educated in travel strategies, it was unsurprising that they were most likely to implement strategies in association with travel that involved crossing a greater number of time zones and travelling in an eastward direction. This study found that athletes and younger participants demonstrated greater flexibility in adjusting their sleep schedules pre-travel. Olympic coaches and support staff have described elevated workload particularly in the pre-Games period[65] and travel requirements are widely reported as a source of work-family conflict for sporting personnel.[65,66] High pre-travel workload and family commitments may limit the capacity of support staff to effectively implement pre-travel strategies.

Study limitations

The findings of this study should be interpreted in consideration of study limitations. First, the data is based on participant retrospective self-report, which may be at risk of errors and bias in retrieval.[67] The surveys did not address the appropriateness of the application of strategies, that is, the timing of light exposure and avoidance is critical for adaptation, and optimal schedules for light exposure depend on the time zones crossed, the origin, the destination and even the season.[68] The surveys were carried out pre-COVID-19 and did not address strategies associated with infection control, which has increased in importance since the pandemic.[32] The study also did not explore the distribution of participants across cabin classes. It has recently been suggested that business class travel preserves sleep duration and quality.[43] The surveys did not include an exhaustive list of symptoms associated with LH travel or potential strategies; however, the questions were open-ended to allow the participants the opportunity to input their own information, outside that captured by the survey. In this study, we asked participants about the symptoms of, and use of strategies for, jet lag and travel fatigue together, and we did not ask participants to distinguish between the two.

Practical implications

The relatively quick recovery times in the present study participants following LH travel suggests good awareness of the negative effects of LH travel and effective implementation of travel strategies. However, this study finds that sleep disruption and fatigue continue to be reported frequently by HP athletes and support staff following LH travel and should be the focus of attention of future strategy development. Special consideration should be given to work and training schedules in the week before travel to minimise sleep disruption and to ensure that athletes and staff embark on their journey in a low-stress and well-rested state.[12] Travel schedules that minimise sleep loss and strategic napping in-flight and on arrival may also be useful as a countermeasure to sleep disruption encountered before, during and after LH travel and daytime fatigue on arrival. There are some situations (such as evening chronotypes) where earlier bedtimes might be recommended to help manage fatigue due to early awakening. Where athletes and support staff are required to perform a short time after LH travel or the stakes for peak performance on arrival are high, it is suggested that organisations should considering upgrading airline seating to promote sleep and help minimise fatigue. This may also be helpful in alleviating stiffness and soreness which is also a prevalent symptom in this population following LH travel. In the case of eastward travel across numerous time zones, HP athletes and support staff may consider seeking professional support and guidelines that have been individualised for their journey and circumstances, which might include pre-travel shifting of circadian rhythms by adjustments to sleep and light exposure schedules.[69] The appropriate ingestion of melatonin under medical supervision may assist this process.[70] It is also recommended that HP

athletes and support staff continue to implement some of the strategies developed during the COVID-19 pandemic to reduce the risk of contracting illness while travelling. To minimise the impact of LH travel on recovery, performance and health of HP athletes and support staff, these study findings suggest that organisational resources should be allocated to minimise pre-travel stress and fatigue caused by athlete training and support staff work programmes, to develop travel strategies that are specific to the individual and journey, and to provide high-quality travel and accommodation that promote sleep and recovery.

To our knowledge, there have been no previous studies which have captured HP athletes' and support staffs' perceptions of LH travel and their use of self-management strategies. In general, LH travel was widely perceived by the study participants to be disruptive to physical and mental performance and increase the risk of illness and injury. The most prevalent symptoms reported following their last LH journey for training or competition were related to fatigue and disrupted sleep schedules. We found that participants perceived themselves to be well-informed with regards self-management strategies for LH travel and implemented multiple strategies before, during and after the flight – the most common were ensuring that they went to bed and ate meals at an appropriate destination time. These current findings suggest that future support and guidelines to athletes and support staff should focus on steps to prevent and alleviate fatigue and sleep disruption following LH travel. Although there was evidence that HP athletes and support staff took the primary steps required to align the body clock to their destination time, very few implemented strategies that have been proposed to be effective in realigning the body clock such as light exposure/avoidance, shifting bedtime pre-travel and melatonin supplementation. More applied research is required to gain a better understanding of the impact of LH travel on the health and performance of HP athletes and support staff travelling for training or competition, and the barriers which prevent the implementation of certain strategies. This will allow the translation of circadian research to better align with the needs and priorities of HP athletes and support staff, and shape and inform future support and guidelines for this community.

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ORCID iDs

Antonia Rossiter https://orcid.org/0000-0002-2284-7072
Alan M. Nevill https://orcid.org/0000-0003-0506-3652

Supplemental material

Supplemental material for this article is available online.

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